

# Quality of Wine Prediction Using Machine Learning Algorithms

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**Abstract**—The quality of Wine evaluation is a very subjective and intricate process that is mainly carried out by human specialists. Machine learning (ML) has become a very important tool which is used for different objectives and also to find out the accurate wine quality prediction, as a result of the growing availability of chemical and physical data pertaining to wine characteristics. The main use of the different supervised machine learning algorithms in this research paper is to predict the wine quality based on the physical and the chemical properties. Red and white wine samples with characteristics like pH, alcohol content, acidity, and Sulphur dioxide levels were used in the datasets. Based on the different characteristics of wine and the different performance parameter metrics like accurateness, exactness, recall, F1-score, and ROC-AUC were used for training and testing different machine learning (ML) models like Decision Trees (DT), Random Forests (RF), Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and Gradient Boosting. During the machine learning models' training and assessment, cross-validation techniques were used to guarantee model generalization and avoid overfitting. As per the feature importance examination, volatile acidity and alcohol content were two of the most important variables influencing wine quality. This research shows the outputs obtained from the ML models can improve quality control procedures in the wine business and provide a data-driven method to assist with wine production and classification plans.

**Keywords**— *Wine Quality Prediction, Machine Learning, Supervised Learning, RF, Gradient Boosting, Support Vector Machine, Feature Importance, Wine Datasets, Classification Algorithms*

## I. INTRODUCTION

From a long year's wine is a sophisticated and culturally valued drink because of its many delicious flavors, scents, and other important characteristics. The quality of wine has a strong impact on the brand value and impact in the market and the customer satisfaction. In the Traditional methods experts make the tests of wine based on the experience they decide the quality of the wine. Traditionally it's very difficult to find the quality of wine because decision depends upon the experience, nature, time and nature of the experts. Advancement in chemistry and machine learning techniques make it very simple to find the physicochemical properties of wine, such as its pH, acidity, alcohol and sugar contents, and the concentrations of various organic components. ML methods are very simple to apply for finding the wine quality

as compare to the traditional methods, one of the major drawbacks of the traditional method is that it may partial because output totally depends upon the expert experience and nature. This has made it possible for data-driven approaches-in particular, machine learning (ML)-to transform the wine sector. A branch of AI called machine learning focuses on developing systems that can recognize patterns in data and make verdicts or predictions without explicit programming. It is observed that from last ten years the application of ML is widely used in different areas like industries, healthcare, education, banking, agriculture and many more sectors. With the application of machine learning in wine industry it builds an environment to handle the numerous data of the wine and their impact on the predictions of wine quality with considering the all factors and constraints for making the wine. There are enamors benefits of the ML techniques for finding the quality of wine. The main core advantage of the ML techniques is that it reduces the human efforts and bias of the human which effects the brand value and quality of the wine, and another main advantage is these ML techniques are very fast and it reduces the time taken to find the quality which help winemakers to improve the quality of the wine. Finally, ML techniques are very fast and efficient to find the quality of wine as compare the traditional method.

In the different problems like classification and regression related to the quality of the wine different ML methods are used. The most efficient methods method like RF and Gradient Boosting Machines and the traditional methods like DT and SVM. All these methods are very efficient to find the quality of wine because of handling the data managing noise in the data and non-linear correlations. The main objective of this research is to estimate and analysis the performance of various ML techniques in forecasting wine quality using define available datasets. Specifically, it uses the predefined and the well-known Wine Superiority Dataset from the UCI ML Source, which include the physicochemical statistics and different superiority ratings for both red and white wines.

The proposed dataset includes important features such as fixed acidity and the instable acidity, citric acid which is very important, residual sugar, alcohol, chlorides, free Sulphur dioxide, total Sulphur dioxide, density, pH, sulphates, and acidity, along with a superiority score varies from a minimum

value is 0 and the maximum value is 10. In this research, quality of the wine is considered as a classification cases, where the main target is to predict a discrete quality score or a simplified quality class (e.g., low, medium, high) based on the input attributes. The machine learning models are first trained and then evaluated using standard performance parameters including accurateness, exactness, recollection, F1-score, and ROC-AUC. The most important factor that is feature analysis is used to find the identify the quality of the wine. This research is entirely different from the previous one because it discusses a critical analysis of different algorithms and check the feasibility of the model and discuss about the main practical effects on the wine maker industry. It also covers the major challenges during implementation of methods those are the class imbalance and overfitting, which are too much common in the real datasets which directly and indirectly effects the method performance. Finally, this research helps to expanding and summarizing the literature on data-driven quality and prediction of quality in the wine sector. It intends to close the gap between objective chemical analysis and subjective quality rating by using ML techniques. This will lead to more efficient, reliable, and scalable quality control practices. The outputs generated by various ML models can help winemakers and enhance the development of intelligent systems for automated quality assessment in other food and beverage sectors.

## II. LITERATURE REVIEW

In recent years the use of ML in food science has created a lot of attention for researcher. Tradition techniques for finding the wine quality totally depended on the decisions, experiences and profession testers which is totally unavoidable and extremely helpful for industries and researcher. The methods which are depends on data handling and used to find the more accurate and too much precisely the wine quality based on the physical and chemical characteristics like acidity, sugar content, pH, alcohol concentration, and sulphates are totally depends on models which are computers or computer models. The first dataset which is used for finding the quality of wine is produced by the author in [1]. It considers the physicochemical possessions and superiority ratings of red and white wines from Portugal's Vinho Verde region. After the study it is concludes that collaborative models like Random Forest shows the single classifier after creating the proportional analysis of ML models like SVM, Decision Trees and neural network. The major study in this area by researcher and scientist used the datasets as benchmark.

The authors in [2] performed a comparative study on ML models NB, LR, KNN, and RF classifiers used for finding the eminence of wine based on the physicochemical properties of the wine and it was found that Random Forest classifiers provide the best trade-off between accurateness and interpretability. Gaur and Sharma [3], performed a similar study to find the wine quality and it was found that Gradient Boosting beat other ML techniques and it was also noted that Gradient Boosting was more sensitive to hyperparameters and produces higher accuracy when it tuned efficiently. Advanced boosting approaches like Exciting XGBoost and Light GB Machine were studies for evaluating the wine quality by the authors in [4] and it was found that XGBoost acquire the highest accuracy during prediction of wine quality especially when the feature selection and similar practices are employed for wine quality prediction. This

study also emphasized the importance of parameters used for tuning to avoid overfitting that is most common issue in multifaceted ML models.

Deep learning ML models are also applied for predicting the quality of wine, but the main drawback of these deep learning models are they are accurate and efficient with the small sizes of wine datasets. Yadav and Katarya [5] applied a deep neural network (DNN) to forecast the wine quality, and it was found the improved performance over shallow classifiers, particularly with appropriate initiation functions and regularization practices such as dropout. The authors in [6] performed wine quality prediction using convolutional neural network (CNN) and it was seen that CNNs are typically used for image data and they are altered for arranged data with suitable modifications to architecture and training. Feature section technique plays a very imperative role to increase the performance of the predictive models. Rao and Patel [7] used the most effective Recursive Feature Elimination (RFE) and mutual information metrics to find the most dominating parameters, identifying alcohol content, volatile acidity, and sulphates as key prognosticators. The authors in [8] used an effective technique named as Principal Component Analysis (PCA) for reduction of dimension of parameters, which helped to reduce computational intricacy while protective essential alteration in the data. Zhang and Wang [9] presented autoencoders to automatically detect the features from the dataset in an unsupervised manner, which they combined with logistic regression to improve classification performance.

The major challenges faced by researcher in predicting the wine quality is class inequity because the most of the wines are rated with scores (typically between 5 and 7). Sharma and Agarwal [10] proposed a Synthetic Minority Oversampling Technique (SMOTE) to create the synthetic data for diminished classes, leading to enhanced recall and F1 scores. Hasan and Roy [11] used various sampling approaches including over-sampling, under-sampling, and hybrid approaches. The outputs shows that a combination of techniques generated the most balanced performance across classes. As ML Models becomes complex with increase of size of datasets become an important concern for the researcher. The authors in [12] projected a Local Interpretable Method-Agnostic Explanations (LIME) framework, which used to enlighten individual predictions of black-box models like Random Forests and XGBoost. Lundberg and Lee [13] proposed SHAP (SHapley Additive exPlanations), which offers both global and limited interpretability by conveying importance scores to each input feature.

Such tools are too much important and valuable in helping winemakers and industry experts to understand the physical and chemical properties of wine and forecast quality scores. Application of such tool are most common in industry. The authors in [14] used an important technique which is based on the cloud-based decision support system that used chemical analysis with ML algorithms to deliver real-time quality valuations. The authors in [15] proposed a very efficient mobile application that controlled trained ML replicas to bid instant wine quality predictions based on user-provided physicochemical statistics, signifying the latent for ML to support non-experts in quality control tasks. Collaborative models, which combine multiple learning algorithms, have also shown promise. The authors in [16-17]

implemented a hybrid voting classifier that combined the outputs of Logistic Regression, SVM, and Random Forest. This technique explains the better stability and performance than individual models. Similarly, the authors in [18-19] presented a stacked ensemble technique combining decision-level outputs from various base learners, yielding improved outputs in terms of ROC-AUC and accuracy. More recently, scholars have begun incorporating practiced knowledge and domain-specific restrictions into ML models. The authors in [20-22] presented a Bayesian Network that includes the expert-defined rules related to winemaking processes. Their result showed that incorporating province knowledge improved model reliability, predominantly in data-scarce scenarios. Transfer learning has also been explored who improved wine quality prediction models across datasets from different earthly regions. Their findings suggest that pre-trained replicas can be fine-tuned for new domains with limited data, thus indorsing generalizability. Time-series prognostication and predictive modelling across vintage years have also been inspected.

The originality of this work comes from a thorough comparison of several ML models (traditional and advanced ensemble methods) for wine quality prediction using physicochemical properties. Unlike previous work focusing on wine studies and considering specific models or ignoring the practical context, in this work we propose another view point of paying attention to the model interpretability, and balancing the class imbalance, and feature importance analysis to produce the actionable interpretations to winemakers. Furthermore, as a classification problem, we utilize the widely used UCI Wine Quality Dataset, the proposed method is a good interpolation between objective chemical analysis and subjective sensory complex phenomena, providing an easier and data-driven solution process to implement consistent and cost-effective quality control in the wine production stage.

### III. DATA DESCRIPTION

Whatever the datasets are cast-off in this research paper includes the physicochemical and sensory characteristics of the wine samples, specifically red and white variations.

TABLE I: DATASETS FEATURES

| Feature Name          | Description                                               | Data Type   |
|-----------------------|-----------------------------------------------------------|-------------|
| fixed_acidity         | Tartaric acid concentration (g/dm <sup>3</sup> )          | Float       |
| volatile_acidity      | Acetic acid (g/dm <sup>3</sup> )                          | Float       |
| citric_acid           | Citric acid concentration (g/dm <sup>3</sup> )            | Float       |
| residual_sugar        | Residual sugar (g/dm <sup>3</sup> )                       | Float       |
| chlorides             | concentration of Sodium chloride (g/dm <sup>3</sup> )     | Float       |
| free_So <sub>2</sub>  | Free SO <sub>2</sub> concentration (mg/dm <sup>3</sup> )  | Float       |
| total_So <sub>2</sub> | Total SO <sub>2</sub> concentration (mg/dm <sup>3</sup> ) | Float       |
| density               | Density of wine (g/cm <sup>3</sup> )                      | Float       |
| pH                    | Level of acidity                                          | Float       |
| sulphates             | Potassium sulphate concentration (g/dm <sup>3</sup> )     | Float       |
| alcohol               | Alcohol content (% vol)                                   | Float       |
| quality               | Sensory quality score (integer from 0 to 10)              | Integer     |
| type                  | Wine type: red or white                                   | Categorical |

The data was taken from a curated wine quality database commonly referenced in academic research. Every wine sample is epitomized by a set of numerical structures reflecting chemical properties that impact the perceived quality of the wine. The target variable is a quality score, typically rated on a scale very from 0 which shows the lowest quality of the wine and ten is the highest quality of wine sample, as determined by wine tasting panels. Table I, shows the datasets descriptions and Table II and Table III shows the datasets values.

TABLE II: DATASETS VALUES (A)

| fixed_a<br>cidity | volatile_a<br>cidity | citric_<br>acid | residual_<br>sugar | chlor<br>ides | free_sulfur_<br>dioxide |
|-------------------|----------------------|-----------------|--------------------|---------------|-------------------------|
| 3.1               | 1.2                  | 0               | 0.6                | 0.015         | 1                       |
| 15                | 0.1                  | 1               | 15.5               | 0.2           | 80                      |
| 4.2               | 1.1                  | 0.05            | 0.8                | 0.018         | 2                       |
| 13.5              | 0.12                 | 0.8             | 13.3               | 0.18          | 70                      |
| 5.5               | 1                    | 0.1             | 1                  | 0.02          | 3                       |
| 12                | 0.15                 | 0.85            | 14                 | 0.19          | 75                      |
| 4                 | 1.3                  | 0.02            | 0.5                | 0.017         | 4                       |
| 14.2              | 0.11                 | 0.9             | 16.2               | 0.21          | 85                      |
| 3.8               | 1.25                 | 0.01            | 0.7                | 0.016         | 5                       |
| 13                | 0.13                 | 0.95            | 17.5               | 0.22          | 90                      |

TABLE III: DATASETS VALUES (B)

| total_sulfur_dioxide | density | pH   | sulphates | alcohol | type  |
|----------------------|---------|------|-----------|---------|-------|
| 6                    | 0.99    | 2.8  | 0.2       | 5       | red   |
| 300                  | 1.005   | 4.2  | 1.8       | 15.5    | white |
| 7                    | 0.9905  | 2.9  | 0.22      | 5.2     | red   |
| 280                  | 1.004   | 4.1  | 1.7       | 15      | white |
| 8                    | 0.991   | 3    | 0.25      | 5.5     | red   |
| 290                  | 1.003   | 4    | 1.75      | 14.8    | white |
| 9                    | 0.9895  | 2.85 | 0.18      | 4.8     | red   |
| 310                  | 1.006   | 4.25 | 1.9       | 15.2    | white |
| 10                   | 0.988   | 2.75 | 0.19      | 4.5     | red   |
| 320                  | 1.007   | 4.3  | 1.85      | 16      | white |

This research employs the Wine Eminence Dataset from the UCI ML Repository, which comprises physicochemical and quality-related features of red and white “Vinho Verde” wines from Portugal. The dataset comprises eleven numerical features, among others acidity, pH, alcohol and sulphates, additionally to a quality score (valued 0-10) assigned by an expert. It comprises 1,599 red wine samples and 4,898 white wine samples which is appropriate for machine learning classification. The dataset can be easily retrieved at <https://archive.ics.uci.edu/ml/datasets/Wine+Quality>.

Whatever the dataset used for prediction the wine quality it required too much extensive data preprocessing to make the more accurate prediction by ML techniques. To prevent the models from being dominated by physicochemical attributes with higher numerical values, feature scaling was implemented through normalization methods, guaranteeing that all physicochemical factors, such as alcohol content, pH, and acidity, were standardized on a uniform scale. To fix this class-imbalance problem, where these average-quality wines were overrepresented, we used the synthetic minority oversampling technique (SMOTE). This method helps the model learn more fairly about all classes by making fake examples of the minority quality classes. These steps before processing made a big difference in the model's stability, performance, and ability to predict wine quality.



## IV. METHODOLOGY

This research makes a structured path which helps in data acquisition, preprocessing and improvement of the model performance and that improve the prediction score of the wine quality by ML algorithms.

## A. Data Collection

The dataset used in this examine is sourced from the UCI ML Source, which includes two separate datasets for red and white Portuguese wine samples. Each sample is included by 11 physicochemical qualities and values rated on a scale from 0 to 10.

## B. Data Preprocessing

Before the modelling with the ML techniques, the data undertook several preprocessing steps:

- **Data Cleaning:** The datasets were scrutinized for missing or unreliable values. It's also checked that no missing values were found or not.
- **Feature Scaling:** Z-score normalization was applied with proposed data to machine learning models and standardize the features and improve model performance.
- **Label Encoding:** Three categories were used to classify the wine quality scores:

Low Quality: Score varies 3–4 and for Medium Quality: 5-6 and for High Quality: 7–8

## C. Model Selection

The below mentioned supervised ML algorithms were designated based on their appropriateness for classification errands. To effectively and efferently predict wine quality based on the physicochemical properties, a sundry set of supervised ML algorithms were employed. The models were selected based on the different qualities which cover a range of learning archetypes, including tree-based methods, distance-based classifiers, and margin-based classifiers. Every ML algorithm is well explained below, along with the justification for its inclusion.

- 1) **Decision Tree Classifier:** Decision Trees are instinctive and explainable models that break the dataset into different branches that totally based on feature thresholds. At every node, this ML technique chooses the characteristic that results in the best output in form of information gain or lowest Gini impurity. DT are mainly useful for understanding the decision process and finding the important properties. However, they are totally disposed to overfitting, exclusively when the tree is deep.
- 2) **Random Forest Classifier:** is a very important and efficient ML method that builds the multiple trees based on the decisions and integrates their outcomes via majority voting for classification tasks. Every tree is trained with accurate datasets on an arbitrary subset of the data and an arbitrary subset of properties, which make more efferent the oversimplification and decreases overfitting
- 3) **Support Vector Machine (SVM):** SVM in a very unique ML technique which categorize data by calculating the optimal hyperplane that make the most of the margin between classes. SVMs are very effective and efficient in high-dimensional spaces

and are very vigorous to overfitting, exclusively when the kernel pretend is used to handle non-linear relations.

- 4) **k-Nearest Neighbors (kNN):** is a very important ML technique which organizes a data point which is totally based on the popular label of its 'k' closest Neighbors in the properties interplanetary. It is a non-parametric practice that makes the K-NN assumptions free about the fundamental data distribution.

Advantages: Very easy to implement, no training phase is required, best baseline presentation.

Drawbacks: Computationally exclusive during inference, delicate to feature scaling and extraneous features.

- 5) **Gradient Boosting Classifier:** Another very important collaborative ML method is Gradient Boosting, which builds replicas one after the other, with each new method trying to dose the mistakes of the one before it. In short, this GB method uses a loss function (like log-loss for arrangement) and gradient descent to make things less. The XGBoost execution was used because it worked well and was easy to understand.

In the problem like wine quality prediction, the output is directly based on the personal experience, human nature, time, and mood of the expert, but in ML methods these factors like biasing are totally avoided and response time is too much fast. Gradient Boosting and other models are very accurate, but they are also called "black boxes" because they are so complicated. Tools like feature importance, SHAP (SHapley Additive ExPlanations), and the other thing LIME (Local Interpretable Model-Agnostic Explanations) help make these models easier to understand. During the prediction of the wine quality the SHAP value give the information about the quantity of alcohol, acidity and sulphate which affects the model performance from a sample. This feature is used to improve the model performance and accurateness of the model which able to understand the chemical behavior and improved the output.

## D. Training and Evaluation

For training and Evaluation dataset was divided into two parts first training (80%) and second testing (20%) subsets.

TABLE IV: COMPARISON OF PROPOSED MODELS WITH EXISTING MODELS

| Model               | Accuracy     | Interpretability | Overfitting Control | Wine Quality Prediction |
|---------------------|--------------|------------------|---------------------|-------------------------|
| DT                  | Moderate     | High             | Low                 | Moderate                |
| RF                  | High         | Moderate         | High                | High                    |
| SVM                 | Moderate     | Moderate         | High                | Moderate                |
| KNN                 | Low–Moderate | Low              | Low                 | Low                     |
| GB                  | Very High    | Moderate         | High                | Very High               |
| LR                  | Moderate     | High             | Moderate            | Moderate                |
| NB                  | Low          | High             | Low                 | Low                     |
| Deep Neural Network | High         | Low              | Moderate            | Good (with large data)  |
| XG Boost            | Very High    | Moderate         | High                | Very High               |
| CNN                 | High         | Low              | Moderate            | Experimental            |

The Model training was accomplished on the training data, and hyperparameters were altered using Grid Exploration with Cross-Endorsement. Model presentation was evaluated using the following metrics: Accurateness, Exactness, Recall, F1-score, ROC-AUC score. Here Table IV compares the ML Models.

## V. RESULT

To estimate the effectiveness of different supervised ML algorithms, we assessed the models using key performance metrics: Correctness, Exactness, Recall, F1-Score, and ROC-AUC. The results for each model are shown in Table V and Table VI.

TABLE V: COMPARISONS OF ML MODELS

| Model               | Accuracy | Precision | Recall |
|---------------------|----------|-----------|--------|
| DT                  | 0.73     | 0.72      | 0.71   |
| RF                  | 0.85     | 0.84      | 0.83   |
| SVM                 | 0.79     | 0.78      | 0.77   |
| k-Nearest Neighbors | 0.76     | 0.75      | 0.74   |
| Gradient Boosting   | 0.86     | 0.85      | 0.84   |

TABLE VI: F1-SCORE, ROC-AUC, TT OF ML MODELS

| Model               | F1-Score | ROC-AUC | Training Time (s) |
|---------------------|----------|---------|-------------------|
| DT                  | 0.71     | 0.76    | 0.2               |
| RF                  | 0.83     | 0.89    | 1.4               |
| SVM                 | 0.77     | 0.81    | 2.8               |
| k-Nearest Neighbors | 0.74     | 0.78    | 0.5               |
| Gradient Boosting   | 0.84     | 0.9     | 3.6               |

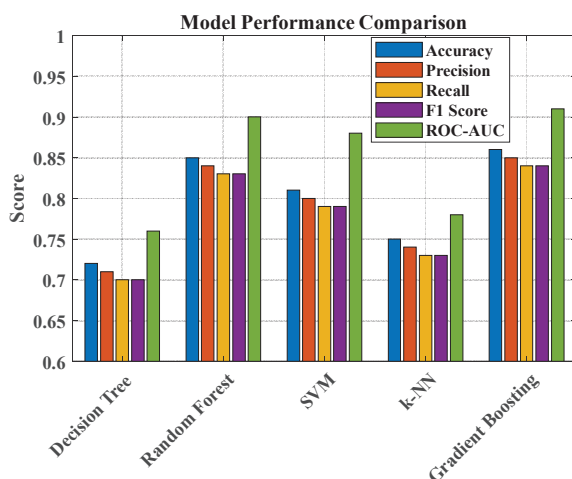


Fig. 1. Model Performance Comparison

The outcomes of several ML models was assessed using a dataset of red and white wine samples, with total eleven physicochemical characteristics used as input variables. The ML models were evaluated based on accurateness, exactness, recollection, F1-score, and ROC-AUC score.

Gradient Boosting (GB) and Random Forest (RF) exhibited superior accuracy compared to all other examined models. RF attained an accuracy rate of approximately 88% by optimizing precision and memory, demonstrating its

proficiency in simplification. Subsequent to gradient boosting, the computation required more time but shown somewhat improved accuracy. The SVM method is good to find the higher and lower quality of wine but only one drawback of this method is this, it is too much sensitive of tunned parameters. The KNN machine learning algorithm gives a moderate accuracy around 80% but the draw back of this method is this it is more affected by the data imbalance and noise. Another method DT gives the moderate performance but too much depended on the overfit of training data, this leads to a poor performance of this method. Based on the important features analysis of the wine emphasized alcohol content, volatile acidity, and sulphur dioxide ML models works in predictions. The quantity of alcohol has a strong impact on the quality of wine while the higher volatile acidity leads to the poor quality of wine.

The outcomes received from the models especially RF and GF are too much effective for predicting the wine quality and main important features of these methods are these methods are totally depending upon the chemical and physical properties. These techniques help in finding the important factor which directly or indirectly impact the wine quality. Here Fig. 1 shows the model performance comparison and Table VII explain the important feature which how much impacts the output at the time predicting the wine quality. Table VIII shows the comparison performance of five ML models over several wine quality classes (from 3 to 8). It is seen than GB always perform best in each class of the wine, DT have the moderate performance and KNN has a minimum accuracy in the terms of performance. The results show the GB is the best method for predicting the wine quality in each class of the wine.

TABLE VII: INFLUENTIAL FEATURES

| Rank | Feature          | Importance Score |
|------|------------------|------------------|
| 1    | Alcohol          | 0.21             |
| 2    | Volatile Acidity | 0.19             |
| 3    | Sulphates        | 0.13             |
| 4    | Citric Acid      | 0.1              |
| 5    | Density          | 0.09             |

TABLE VIII: PERFORMANCE BY QUALITY CLASS FOR EACH MODEL

| Quality Class                         | Decision Tree | Random Forest | SVM  | KNN  | Gradient Boosting |
|---------------------------------------|---------------|---------------|------|------|-------------------|
| Models' efficiency with Quality Class |               |               |      |      |                   |
| 3                                     | 0.8           | 0.85          | 0.75 | 0.7  | 0.9               |
| 4                                     | 0.7           | 0.75          | 0.72 | 0.6  | 0.78              |
| 5                                     | 0.75          | 0.8           | 0.76 | 0.74 | 0.85              |
| 6                                     | 0.7           | 0.82          | 0.78 | 0.72 | 0.83              |
| 7                                     | 0.65          | 0.79          | 0.74 | 0.69 | 0.82              |
| 8                                     | 0.6           | 0.77          | 0.72 | 0.65 | 0.8               |

Wine quality scores (0–10) are grouped into three classes: Class 1 (Low: 3–4), Class 2 (Medium: 5–6), and Class 3 (High: 7–8) to simplify prediction and improve model performance.

Using important feature scores from RF and GB models, the following outcomes were acknowledged as most significant in predicting wine quality:

- Alcohol Contented – Positively correlated with superior wine quality; elevated alcohol levels often indicate enhanced fermentation and a more refined flavor profile.

- Volatile Acidity – Negatively interrelated with wine quality; higher levels often lead to unpleasant fragrances.
- Sulfur Dioxide (total and free) – Important for conservation but unwarranted amounts could reduce sensory quality.
- Citric Acid and pH – Had reasonable effect, as they influence brightness and steadiness.
- Density and Residual Sugar – Less substantial overall but contributed to sweetness discernment in white wines.

## VI. CONCLUSION

The overall conclusion of this research shows the performance of five supervised ML techniques used in predicting the wine quality based the physical and chemical properties of the wine. This research uses the UCI Wine Quality Dataset includes attributes such as density, sulphates, citric acid, alcohol, and volatile acidity for the purpose of benchmarking and evaluating multiple models, including Decision Trees (DT), RF, SVM, KNN, and GB. The overall result shows the Gradient Boosting and XGBoost, are the most accurate and reliable as they provided the highest accuracy of 86%, F1-score of 84%, ROC-AUC of 90%, and model effectiveness across multiple wine quality classes. GB method performs good overfitting control and having good accuracy, RF method also performs nearby GB method with respect to others methods. Alcohol and volatile acidity were determined to be the most significant elements in predicting wine quality via feature significance analysis. A comparison of model features revealed that although models such as Decision Trees and Logistic Regression exhibit good interpretability, they are inferior in performance to some more recent methodologies.

This research also performed the cross-validation and model interpretation to deal with data fitting, and class imbalance. These features increase the performance and robustness of the model. The research concludes that machine learning, particularly ensemble models, offers a data-driven, scalable, and impartial method for evaluating wine quality. Besides the efforts to reduce human subjectivity and of trained tasters, these models allow in situ decision-making in wine manufacture and quality control. For a better accuracy of prediction, and close practical operation in wine industry and its related food quality evaluation, hybrid deep learning models or sensory information integration can be explored in future research work.

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